# Atharva Rathi

+91 9370877347 | atharvarathi09@gmail.com | atharvarathi.com | linkedin.com/in/atharvarathi | github.com/AtharvaRathi

## Education

**K.J Somaiya College of Engineering** Aug 2023 – Present
*B.Tech in Information Technology — Honours in Artificial Intelligence* *Mumbai, Maharashtra*

## Technical Skills

**Languages & UI/Data**: Python, C++, Go, Java, JS/TS, SQL, React.js, Material-UI, Streamlit, D3.js

**AI, ML & Vision**: PyTorch, Scikit-Learn, XGBoost, YOLOv4/v8, ByteTrack, OpenCV, PaddleOCR, LangChain, LangGraph, CrewAI, RAG, LLMs/VLMs (Ollama, Qwen-VL, CLIP, SigLIP)

**Backend & DBs**: Node.js, Express, FastAPI, Flask, REST, WebSockets, SSE, Microservices, Go Concurrency, AST, PostgreSQL, MongoDB, Redis, FAISS, Qdrant, Weaviate, Pinecone

**Cloud, Edge & DevOps**: Docker, Kubernetes, AWS, Linux, Git, TensorRT, CUDA, Apple MPS, GStreamer, FFmpeg, HashiCorp Vault, OpenSSL

## Experience

**Software Engineering Intern | EmerTech Innovations** Jun 2026 – Jul 2026

- Engineered an automated Digital Product Passport (DPP) ingestion pipeline, accelerating supplier onboarding by eliminating manual data entry bottlenecks.
- Designed a scalable workflow to parse unstructured supplier PDFs, automatically extracting, validating, and structuring critical data into a database-ready schema to flag missing essential fields.

**Software Engineering Intern | Trueview** Dec 2025 – Jan 2026

- Architected a Retrieval-Augmented Generation (RAG) knowledge base using semantic search and vector embeddings, reducing query resolution time for complex CCTV specifications.
- Deployed a scalable AI backend with secure API endpoints and integrated conversational interfaces, aligning LLM capabilities with manufacturing workflows to automate enterprise knowledge-sharing.

**Software Engineering Intern | Manomay** May 2025 – Jul 2025

- Engineered a scalable full-stack e-commerce platform from scratch, driving a 20% increase in Q1 online sales by aligning frontend conversion optimization with robust backend architecture.
- Implemented secure authentication flows and a real-time notification system, optimizing database querying to ensure low-latency performance and a seamless user experience.

## Projects

**SemanticEdge: Edge-to-Cloud Video Analytics Pipeline** | *Python, C++, PyTorch* Link

- Engineered a hardware-agnostic edge pipeline reducing cloud bandwidth by 85% by developing a C++ context firewall that calculates cosine similarity to drop routine visual events.
- Implemented open-vocabulary RAG for zero-shot video retrieval by integrating YOLOv8, ByteTrack, and SigLIP to generate 512-d visual embeddings stored in Qdrant.

**Fault-Tolerant Webhook Dispatch Engine** | *Go, Redis, Docker, Concurrency* Link

- Architected an asynchronous event-delivery system in Go, utilizing worker pools and channels to concurrently process and dispatch thousands of high-volume payloads.
- Engineered a reliable message-queuing architecture using Redis, implementing exponential backoff and a dead-letter queue (DLQ) to ensure system stability against failing external endpoints.

**VStream: Vision-Language Inference Engine** | *Python, PyTorch, Qwen-VL, WebSockets* Link

- Built a token-pruning inference wrapper for Vision-Language Models (VLMs), achieving 15-20 FPS on consumer GPUs via gradient checkpointing and 4-bit quantization.
- Developed a WebSocket-based alerting layer with sub-500ms latency, dynamically adjusting token pruning thresholds based on real-time GPU memory pressure.

**Adaptive Corrective & Self-Reflective RAG** | *Python, LangGraph, FAISS, LLMs* Link

- Built an agentic RAG orchestrator integrating Corrective and Self-RAG frameworks to dynamically select retrieval strategies (dense/sparse) based on query complexity.
- Implemented iterative evidence verification and confidence-based stopping to optimize latency and context size without degrading answer faithfulness.